

BUCKINGHAMSHIRE
NEW UNIVERSITY

EST. 1891



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FY006 - Digital Media
CourseWork 2
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Introduction

I have always considered myself somewhat of an artist, having an eye for detail, finding beauty where others do not, possessing a keen desire to express myself and my ideas. I have favored more traditional methods of expression such as drawing, painting, motion, and music. However, technological advancements have significantly expanded our ability to express ourselves more thoroughly, not just with imagery, sculptures, movement, or music but with a new immersive experience, one that envelops all of the senses, *truly evoking emotion.*

Imagine creating a world, a world of your own design, a world that holds a story—one that your audience engages with as a lived experience. What world would you create? What story would you tell? What ideas would you share in a realm that is only limited by skill and imagination? These questions have led me to this assignment where I have been tasked with keeping a record of my work as I progress through the course FY006 Digital Media, detailing what I have learned, reporting interesting findings during independent research, and displaying my practical abilities.

Week 1 - Digital Images and composition

Blue Pencil Summary



Digital images consist of pixels which are small units that form an image on-screen. The density of pixels affects the screen resolution, and more pixels usually mean a clearer picture. The two main types of digital images are bitmaps, which are more suitable for photos and videos, and the other type are vectors which are ideal for logos and graphic designs. Bitmaps depend on resolution and might not scale well when the image is stretched multiple times, whilst vectors are 'resolution independent', making them scalable without losing picture clarity.

The term bit depth is used to refer to the number of colours in an image, with higher bit depth allowing for more colours. E.g, 8-bit depth means 256 colours, 16-bit depth means 65,536 colours, and 24-bit depth means a vast 16,777,216 colours.

Key Points



Bitmaps may get blurry when resized, like paintings.

Vectors stay clear regardless of size changes, like drawings.

Image resolution is about pixel density, impacting sharpness.

Bit depth influences colour variety, with higher bit depth providing more vibrant colours.

Independent research

The first digital photo

Before the internet age, innovators in digital photography were finding new ways to translate information into visual images. In the early 20th century, experiments in 'electrographic photography' wanted to create images transmittable by electricity. A significant development occurred in 1957 when Russell Kirsch used an early computer to convert an analogue photograph of his infant son into a small digital file (right). (*The first digital photos*, 2015)

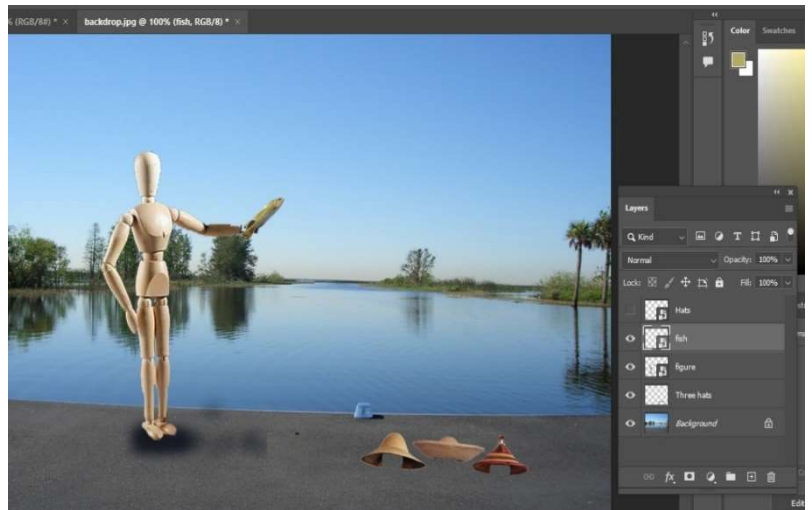


While other researchers focused on converting existing images into a digital format, Peter Noble in 1968 created an 'Active Pixel Sensor' (*Active pixel sensor*, 2022) capable of directly translating light into digital information. This sensor removed the need to use analogue images, enabling the direct creation of digital images straight from real life.

By 1973, Steve Sasson, a Kodak researcher, further advanced this concept by developing the first fully digital camera, capable of capturing and storing electronic photographs. Originally designed to be a handheld device, the resulting camera turned out to be a very sizable machine, roughly the size of a toaster (*Kodak's first digital moment*, 2015). Despite its bulk, it marked a milestone by capturing small black and white digital photographs, 0.8 megapixels to be exact, and saved onto a cassette.

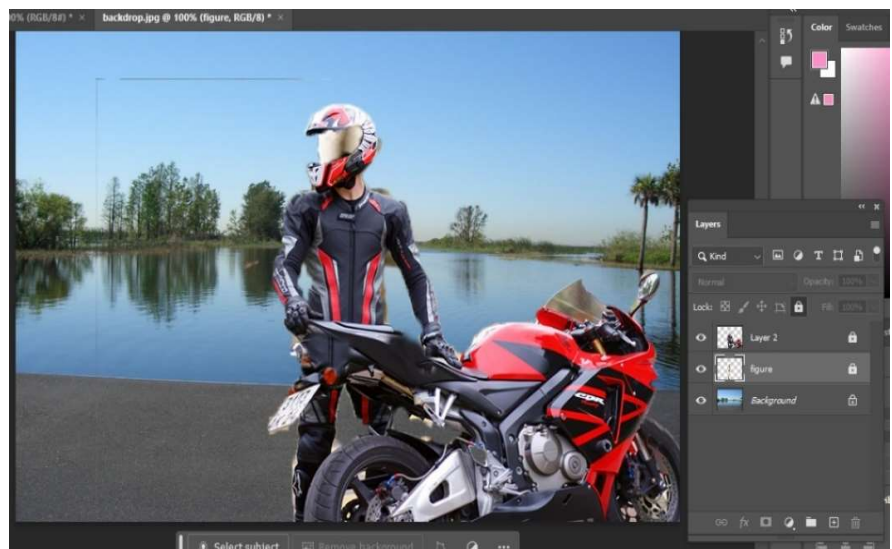
Practical - Compositions

For this practical assessment I have been tasked with creating a composite image using adobe photoshop, dressing a model figure and adding outsourced imagery. The university gave me some starting images to work with so I decided to load them up and see what I can do. Placing the layer in the correct order is vital when creating composite images.



I wanted to dress the figure in a motorcycle suit and use a motorcycle as the outsourced image, then I wondered if perhaps it would be simpler to use the object selection tool to cut the suit and bike out and fit it around the figure. I used the selection tools to select the suit and bike, using the 'layer via cut' option to detach the selected area into its own layer. Then dragged said layer over to the figure image.

The figure model was a little too big to fit the suit as was, and its stance also did not work for the idea. I used the tools at hand to free transform the model, changing its shape and size until I was satisfied with the model's placement. I then used the eraser tool to trim off any of the model that was protruding from behind the suit. Using the model and the backdrop provided, an outsourced image, object selection and layer cutting program functions, I was able to create a Digital composite image.



Week 2- Compression and image extractions

Blue Pencil Summary

Compression is the process that transforms data into a format requiring fewer bits. It can be lossy, which means some of the original data is lost, or lossless, preserving all data bits. The process of compression involves two stages, compression and decompression, which are facilitated by a CODEC (Coder-Decoder). CODECs can be symmetric or asymmetric, with asymmetric compression takes longer to decompress.

Lossy compression sacrifices some data for higher compression in size, while lossless compression keeps all data without losing any. Truncation which is a lossy compression method, takes away the least significant bits from each byte. The GIF format uses indexing to reduce file sizes, limiting the number of colors to 256 and using dithering to create the illusion of extra colors. Examples of lossless compression methods include Run-Length Encoding (.bmp or .tiff) and LZW (mainly .tiff formats).

In summary, file size reduction is achieved through compression methods, CODECs apply both lossy and lossless techniques, and different methods get different results.

Key Points:

Compression is a data transformation process.

It can be either lossy (some original data is lost) or lossless (preserving all data bits).

These stages are facilitated by CODECs (Coder-Decoders).

Truncation, a lossy compression method, removes the least significant bits.

Independent research

Zip File Compression

The creator of the .ZIP file format is Phil Katz, from a company called PKWARE, who developed it in response to legal action directed from Systems Enhancement Associates (SEA). SEA claimed that Katz's archiving products were based on their ARC archiving system. To emphasize the speed superiority of their product over ARC, Katz and his associate Robert Mahoney coined the term "zip," meaning "to move at high speed." The first release of the .ZIP File Format Specification, named APPNOTE.TXT, was introduced in 1989 as part of the PKZIP 0.9 package (Knowledgebase, no date).

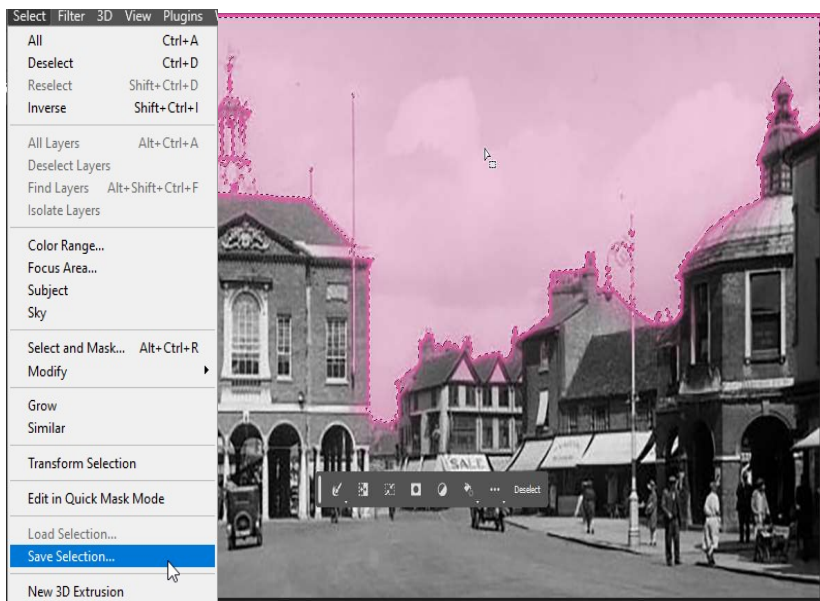
The compression process uses algorithms to eliminate redundancy, optimising the storage of data. The functionality of Zip files involves compression during file creation and decompression during extraction, often facilitated by compression tools or software similar to the program winRAR. (PKWARE, 1989).

The benefits of Zip files include efficient storage, transmission of data, reduced file size, and simplified file organization. I was surprised to find such an indepth exploration of the technical aspects of Zip files, their applications, and the advantages they offer in various contexts.



Practical – Selections, extractions and masks

For this exercise I am going to use the material provided by the university. The image is a picture taken many years ago in High Wycombe. I will show how I use photoshop to make selections, extractions and apply masks to an image.



I used the magic wand tool to quickly and efficiently select the sky portion of the image. Using select and save selection I can edit without worry of losing the selection path. I created a new layer to edit using layer to copy right click option and named this new layer 'mask edit'. Now I will add a colour mask.

Selecting the extracted masked layer, I was able to add some colourisation to the image using the Hue/Saturation option, selecting colourise and adjusting the sliders until I found a nice blue. I believe I have successfully created an image using selection, extraction, masking and colourisation.



Week 3 – Digitising Media and Colourising B&W

Blue Pencil Summary:

Digitisation is the process of converting physical or analogue information into digital (numerical) form. This involves capturing, converting (through sampling and quantising), and storing data. The effectiveness of digitisation depends on factors such as sample rate and bit depth, where higher values lead to a more faithful representation. Digital-to-Analogue Conversion (DAC) involves fetching, converting, and delivering stored digital information, with the term 'digital' not necessarily implying high quality. While digitised representations are never perfect replicas, the goal is to capture as much information from the original as possible. Digital information's versatility lies in its ability to be processed on the same type of machine. In summary, digitised information is an imperfect replica, and accuracy improves with increased sample rate and bit depth, often dictated by system limitations and cost. The devices involved in this process are known as ADCs (Analogue-to-Digital Converters) and DACs (Digital-to-Analogue Converters).

Key Points:

The term 'digital' does not necessarily mean high quality.
 DAC encompasses fetching, converting, and delivering stored digital information.
 The process includes capturing, converting through sampling and quantising, and storing data.
 The effectiveness of digitisation is influenced by factors such as sample rate and bit depth.
 Higher values of sample rate and bit depth contribute to a more faithful representation of the original information.

Independent research

The danger of fragmented processes

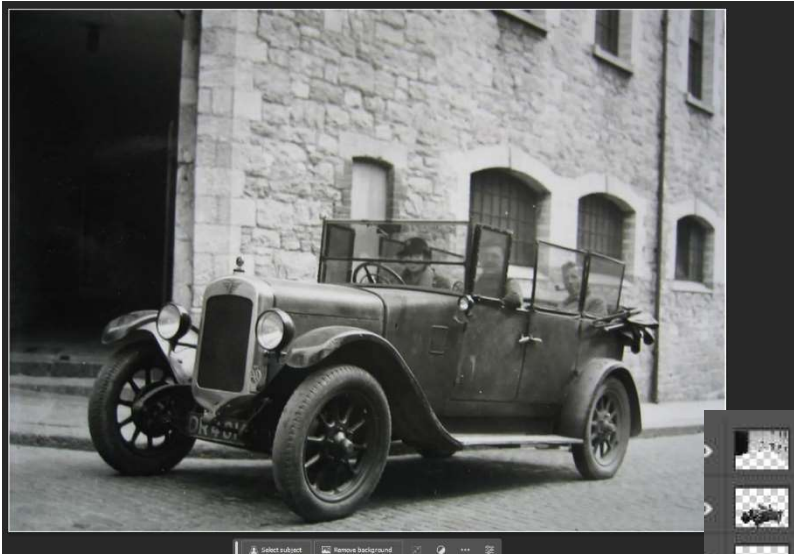
The danger of fragmented processes in digitisation brings about significant security risks, particularly in an already challenging cybersecurity field. A Ponemon Research survey found that over 80% of respondents believed they had experienced data breaches as a consequence of digital transformation initiatives (Nok Nok News, 2022).

Incomplete digitisation efforts contribute to process fragmentation, maintaining manual inputs and reliance on paper based documents. This disorganisation creates vulnerabilities that attackers can exploit, posing risks to sensitive assets involved in purchasing processes, such as financial transactions and personal data.

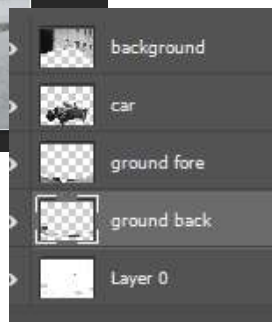


Additionally, the potential introduction of ransomware or other malware further heightens the threat to digitalised assets. In the era of advanced technology, the ability to document forgery increases, presenting a risk of producing convincing IDs and assets that may deceive individuals responsible for manual verification. (Proof, 2023)

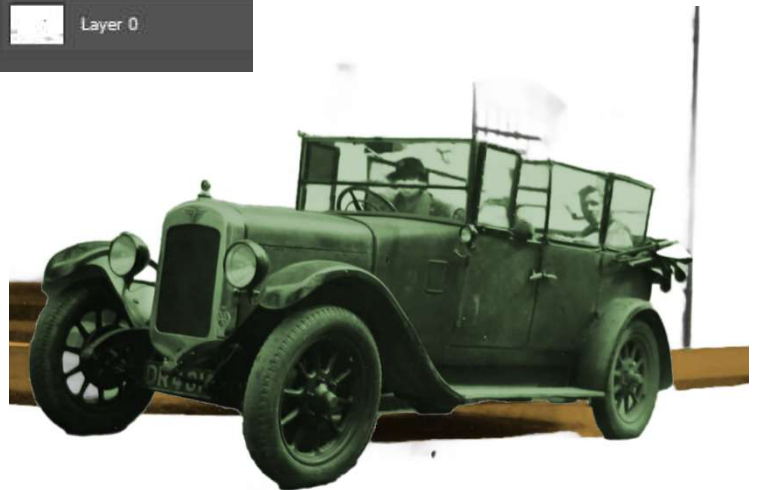
Practical – Colourising B&W



Using the OldCar.PNG supplied by the course material, I will use the skills acquired in the previous weeks to completely colourise this image from black and white.



Selecting, extracted, masking layers I am able to separate the layers from vision using the layer view eye allowing me to better focus on each section. Then using hue/saturation I was able to add some colorisation to the individual layers.



In addition to the Hue/saturation options I added further adjustment using the curve tool, allowing to find precise shades of colour. With more time, skill and effort I see how it will be possible to colour the black and white image as if it was a colour original.

Task completed successfully.

Week 4 - Digital image file types and animation

Blue Pencil Summary:

This summary provides insights into various image file formats, detailing their characteristics and use cases. JPEGs excel in preserving photo quality and are commonly favored in digital cameras. BMPs, a Microsoft made format lacks compression, transparency, and animation features. PNGs, introduced in 1996, are valued for their true color and transparency support, making them ideal for web applications. GIFs, widely used for simple graphics and animations, are limited to 256 colors. Animation is explained as the rapid display of consecutive images, creating a visual sequence. The process involves key frames and inbetweening, essential for making intermediary frames to act as a story board during the animation process. In conclusion, diverse image file types have specific purposes, animated GIFs are most popular for web content, and Adobe Photoshop aids in the editing of dynamic images.

Key Points:

PEG Format: Suitable for photographs, commonly integrated into digital cameras.

Developed by the Joint Photographic Experts Group.

BMP Format: A Microsoft proprietary format.

Lacks compression, transparency, and animation features.

PNG Format: Introduced in 1996, ideal for web applications.

Supports true colour and transparency; APNG offers enhanced animated capabilities.

GIF Format:

Commonly used for simple graphics and animations.

Limited to 256 colours but supports transparency and animation.

Independent research

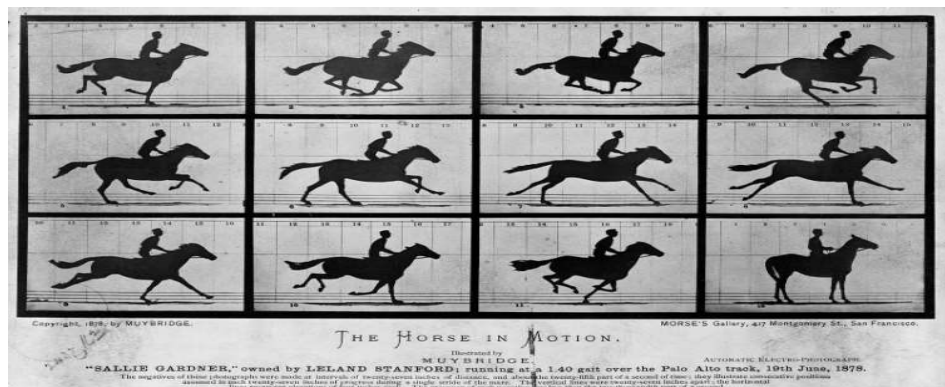
The Horse in Motion

(Pilcher, 2021)

The very first motion picture can be traced back to 1878 when Leland Stanford, a horse breeder, hired photographer Eadweard Muybridge to address the question of whether a running horse lifts all

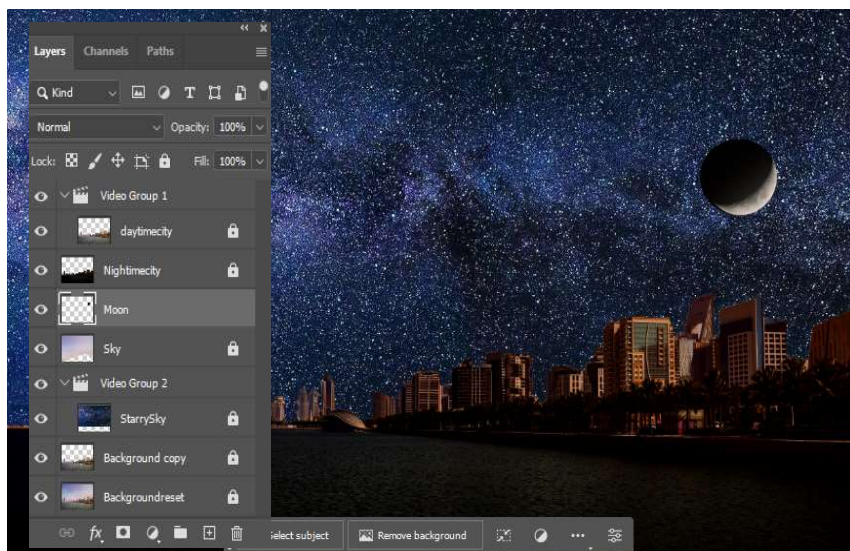
four hooves off the ground at the same time. The first attempt to visually capture this took place in 1873 with Stanford's trotter, Occident. Dark and blurry images were produced leaving Eadweard unsatisfied with the results.

Muybridge spent five years refining his technique and in 1878, he successfully captured the movement of Stanford's racehorse, Sallie Gardner, using 24 cameras arranged side by side. This groundbreaking method, resembling a stop-motion film, provided clear images, confirming Stanford's theory that all four hooves leave the ground when a horse runs. This achievement marked a pivotal moment in the evolution of motion pictures.



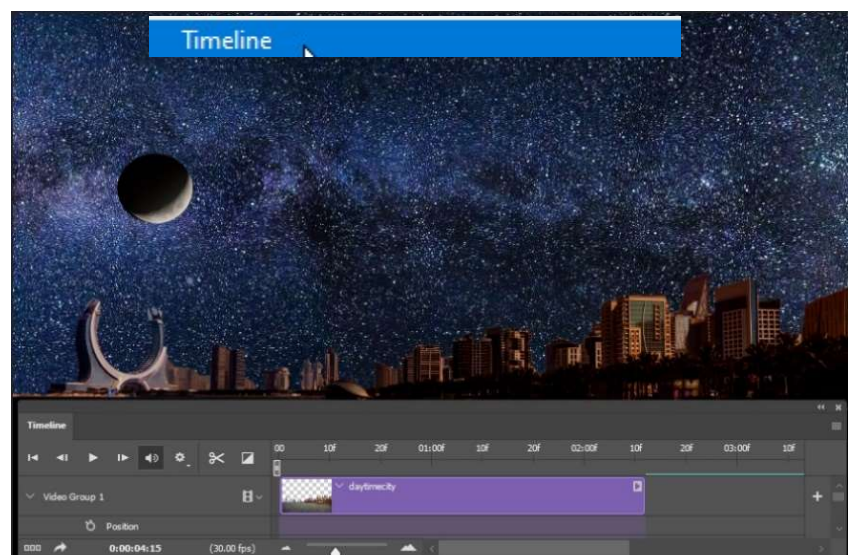
Practical – Animation with Photoshop

In this exercise I will use photoshop to create an animation. The animation I will create is what I would like to use in my first coursework assignment for this module. I will attempt to make the animation appear as if the clouds are moving and time in passing into the evening.



This idea turned out to be slightly more complex than I had first imagined. However, I feel it has bolstered my skill through determination and necessity. I had to create 3 versions of the image to be able to have the animation flow how I wanted. A day slide, an intermediary slide and a finishing night slide.

Using the timeline tool in photoshop I believe I have successfully created the animation how I wanted. Sometimes having a grand idea will drive us to finding way to accomplish these feats. My need for expression ensured that I completed the task in the way I envisioned, advancing me the completion of this task,



Week 5 - Visual Hooks

Blue Pencil Summary:

Creating a visual hook is a skill used to capture the audience's attention. Various techniques like illusions, composition, colours, and animation contribute to creating impactful visuals. Knowing ideas like the 'Rule of Thirds' or the Fibonacci Spiral aids in achieving good visual composition by strategically positioning and arranging objects in a scene or through thoughtful cropping and editing. Understanding colour modes is essential knowledge to have, printed graphics commonly use CMYK (Cyan, Yellow, Magenta, and Black), while screen display graphics often utilise RGB (Red, Blue, and Green). Additionally, spot coloring or colourising monochrome images can be an impactful tool, adding vibrancy and visual appeal where needed.

Key Points:

Developing the skill of creating a visual hook is essential for capturing audience attention. Understanding concepts like the 'Rule of Thirds' or the Fibonacci Spiral is crucial. Various techniques include illusions, composition, colours, and animation.

Independent research

The Fibonacci Spiral

The Fibonacci sequence, conceived by Italian mathematician Leonardo Fibonacci in the 13th century, is a series of numbers with the next number being the sum of the previous two. While lacking any specific formula, the sequence reveals intriguing connections between its numbers (Mitchell, 2023). One such connection is the golden ratio, approximately 1.618, which comes from the division of Fibonacci sequence numbers. This ratio, significant to mathematicians, scientists, and naturalists, even in nature's dimensional properties, as seen in phenomena like beehives, shells and sunflowers. The golden ratio also influences artistic and architectural domains, heavily featuring in structures like the Parthenon in Athens and the Great Pyramid in Giza (Leary, 2019). The Fibonacci sequence, a timeless mathematical concept, unveils inherent patterns and connections in both the natural and human-made worlds.



Practical – Spot Colours

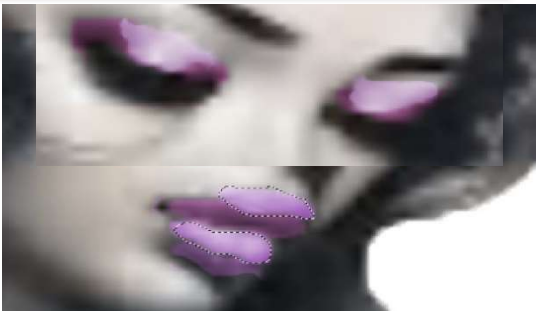


Sakura Cherry Blossoms [EDITED]
Color Palette



Colors in Palette

Color	Hex	RGB
	#bf8065	(191,128,101)
	#541f1f	(84,31,31)
	#901f3b	(144,31,59)
	#ff92c4	(255,146,196)
	#fbb7dd	(251,183,221)



In order to get something that resembles shading on the lips and eyes I used the selection tools to firstly effect the darker areas of the lips and eyes, adding the darker tones, the selecting the lighter areas and continuing the process. Close up the colours don't seem to lay well together but as we have learnt, the human brain makes many adjusts to what we see, fixing a lot of the visual issue and using shortcuts, in turn making the whole image blend together nicely.



Here is the final image using spot colour to add vibrance and attention with a visual hook. I am happy with the way tis has come out. Very close to the idea in my mind. I know with more practice and more research into design ideologies like the golden ratio, I will be able to create even more visually striking art for all to enjoy.

Week 6 – Visual Hooks – Illusions

Blue Pencil Summary:



Our brains tend to take shortcuts to deal with the overload of information, sometimes causing confusion. Digital media cleverly uses tricks to grab our attention by exploiting these shortcuts. We resort to mental shortcuts due to lack of time, complexity, choosing to ignore, being selfish, or simply wanting happiness. In sound tricks, digital media can separate sounds and visuals, creating new and interesting associations using techniques like speeding up, adjusting frequencies, adding echo, or even creating entirely new sounds. Similarly, optical illusions, especially those 3D pictures, are used in ads and effects to create eye-catching and universally understandable visuals.

Key Points:

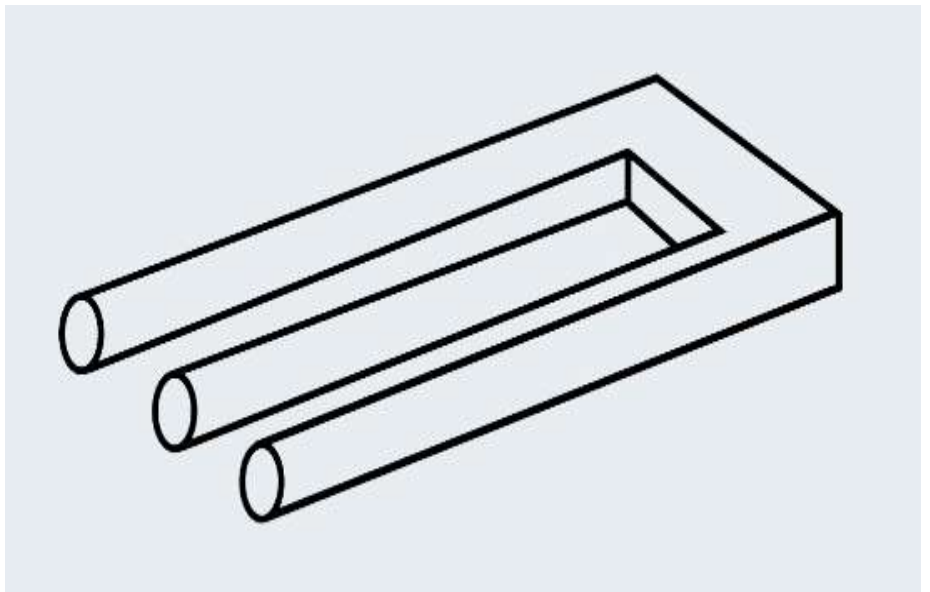


Our brains naturally take shortcuts in processing information, often unnoticed. Digital media uses tricks to capture attention by these mental shortcuts, utilizing various strategies.

Independent research

The impossible Trident

The impossible trident, also known as the impossible fork or blivet, is an optical illusion that creates the perception of a three-pronged object despite only having two prongs. Originating in the 1960s, it gained prominence as a visual puzzle and was further popularised by psychologist Lionel Penrose and mathematician Roger Penrose in their studies on visual perception and optical



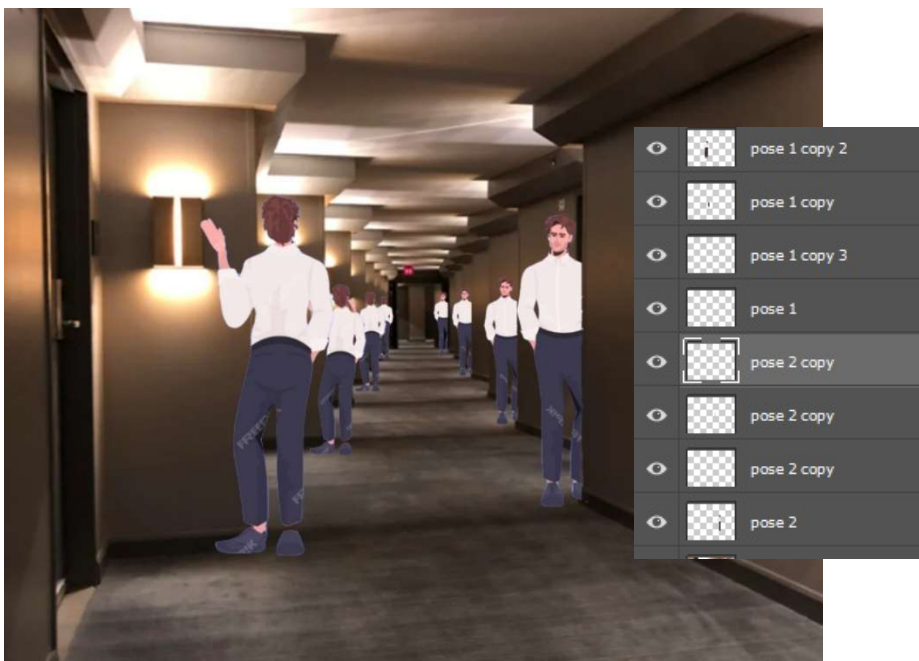
illusions. This intriguing illusion has found applications in various fields, such as art and graphic design, showcasing the ongoing fascination with visual perception and the ways illusions challenge our understanding of reality. (Impossible world, no date)

Practical – Illusions



I found this image interesting to use as material for my illusion creation task as this image looks like the infinite mirror illusion, however, it is simply designed to look that way, in other words, this image contains no mirrors. I like this image so much that I decided to put an illusion in an image that's illusion is that there is no illusion.

The idea was to locate an image of one same character in the same pose but from different angles. I decided to use this image as it is close to the idea. In order to make the idea work I used Photoshop, utilizing the selection and extraction tools to individualise each pose.



This is the final image, I enjoyed playing with the idea of an illusion. Though the poses I found and used were very helpful, editing was still needed. I used free transform to flip the poses and the eraser tool to have the image as correct as possible while creating the illusion of a mirrored image protruding from the wall.

Week 8 - working with Digital Audio

Blue Pencil Summary:

Sound is a type of energy that travels through various materials, with its speed depending on the medium. The pitch of a sound, whether it's high or low, is determined by its frequency, measured in Hertz (Hz). Higher frequencies result in higher pitches, while lower frequencies produce lower pitches.

The intensity or loudness of a sound is measured in Decibels (dB) on a logarithmic scale called the Sound Pressure Level (SPL). Human hearing is limited to a range of frequencies, typically from 30Hz to 20,000Hz. Certain frequencies are more easily perceived by our ears, and equalisation helps balance the volume of different frequencies.

Various audio file types, such as .wav (lossless, uncompressed) and .mp3 (lossy, compressed), differ in size and sound quality. Binaural sound, employed in technologies like Virtual Reality, replicates the way human ears perceive sound by using specialized microphones, enabling the creation of unique audio effects and illusions.

Key Points:

The pitch of a sound, determined by its frequency measured in Hertz (Hz).

Human hearing is limited to a frequency range of about 30Hz to 20,000Hz.

The speed the sound carries varies depending on the medium.

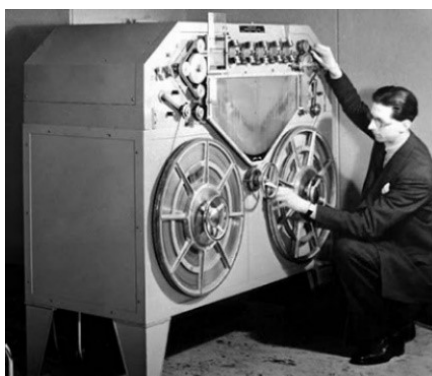
Independent research

Evolution of Digital Audio

The origins of digital audio trace back to Alec Reeves' invention of Pulse Code Modulation in the 1940s. Despite the conceptual breakthrough, practical implementation faced challenges due to the complexity of creating converters with acceptable sound quality. Additionally, the data rate from audio converters exceeded the recording technology capabilities of that era. (Bridge, 2020)

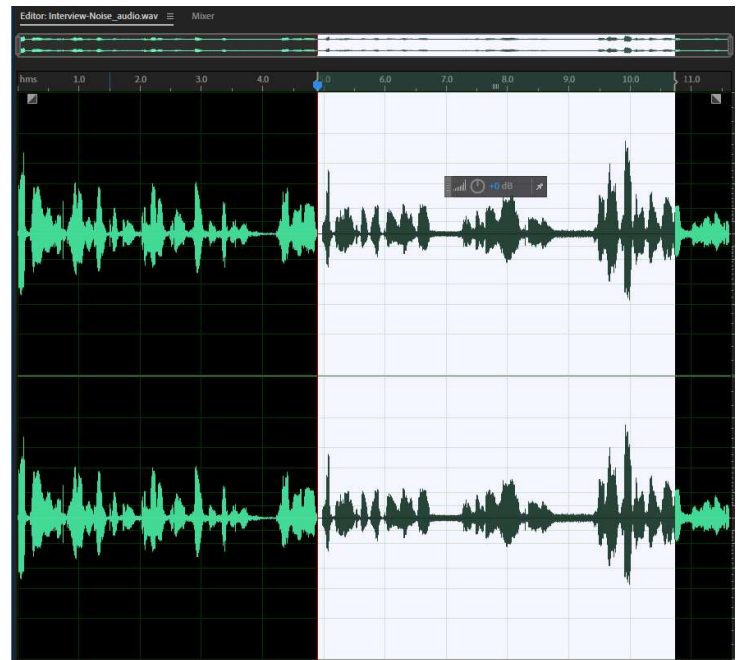
In contrast to the current capacity of IT systems to handle extensive data, early digital audio recording formats like CD and DAT surpassed computer recording capabilities.

This led to the adaptation of audio recording formats for data storage purposes by computers. The first of these was built experimentally by NHK and demonstrated in mono form in 1967. This used a quadruplex tape deck running 14-inch reels of 2-inch wide tape weighing 10Kg. The stereo version was demonstrated in 1969. Denon produced a system based on the NHK machine for vinyl disk mastering in 1972. The machine weighed 400Kg.

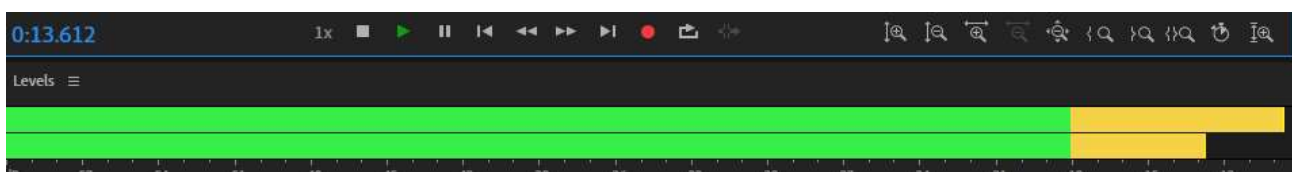
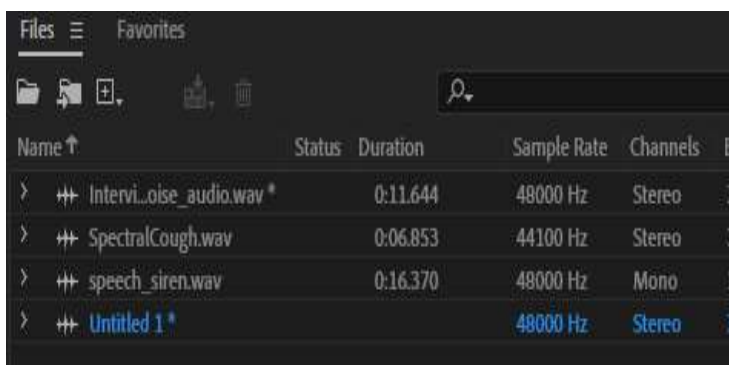


Practical – Editing Audio

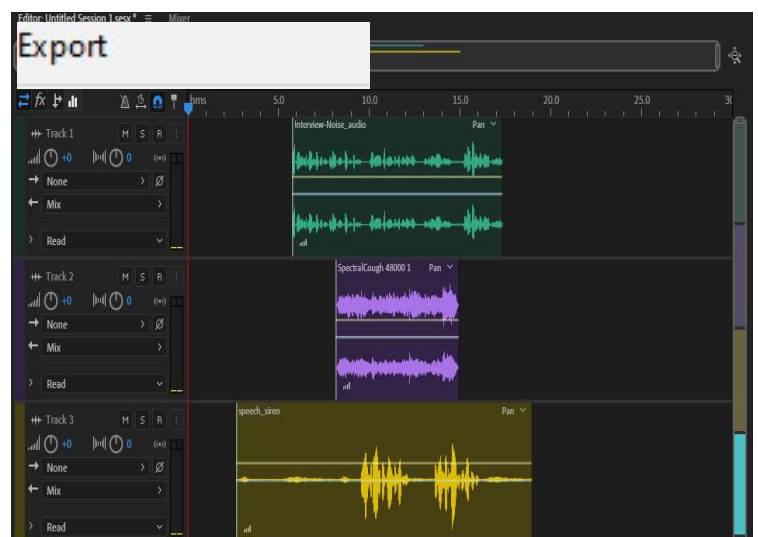
Adobe audition is something I have absolutely no experience in, so I was very looking forward to learning how to edit sound and music. Music being one of the purest forms of expression, I intended to absorb as much information on this program as possible.



Like the other programs I had little to no experience with, 'playing around' with the settings, option and tools helped me navigate the program and find new and interesting ways to apply change to the sounds. I added multiple tracks of sounds in order to create something unique. Using the record and sound tools I was able to make some viable recordings and edits.



Spending lots of time 'playing around' with the program I believe I have successfully completed the task of digitally editing sound. I will explore this program in more depth in my own time, but for now its time to export the file to .MP3 and move on.



Week 9 - 3D Printing

Blue Pencil Summary:

3D printing begins with the creation of a computer-generated model using Computer-Aided Design (CAD). A specific software, called a slicer, processes the model into thin horizontal layers. These layers are then transmitted to the 3D printer, which reconstructs the model layer by layer. Various 3D printer types, such as Fused Filament Manufacturing (FFM), Power and Glue, Stereo Lithography Apparatus (SLA), and Selective Laser Sintering (SLS), utilize different materials and methods.

However, even with this groundbreaking technological marvel, 3D printing still comes with its hindrances. Layer thickness becomes a common issue, like the canvas texture influencing an artist's brushstroke. In FFM 3D printing, the nozzle diameter is like the tip of a fine pen for a calligrapher, great detail and precision are required. Layers, while integral, create subtle weaknesses along their seams, much like the of stitching in fabric, leaving 3d printed objects every so slightly structural unstable, causing possible ramifications on the object's longevity.

Key Points:

3D printing starts with computer-generated models created through Computer-Aided Design (CAD).

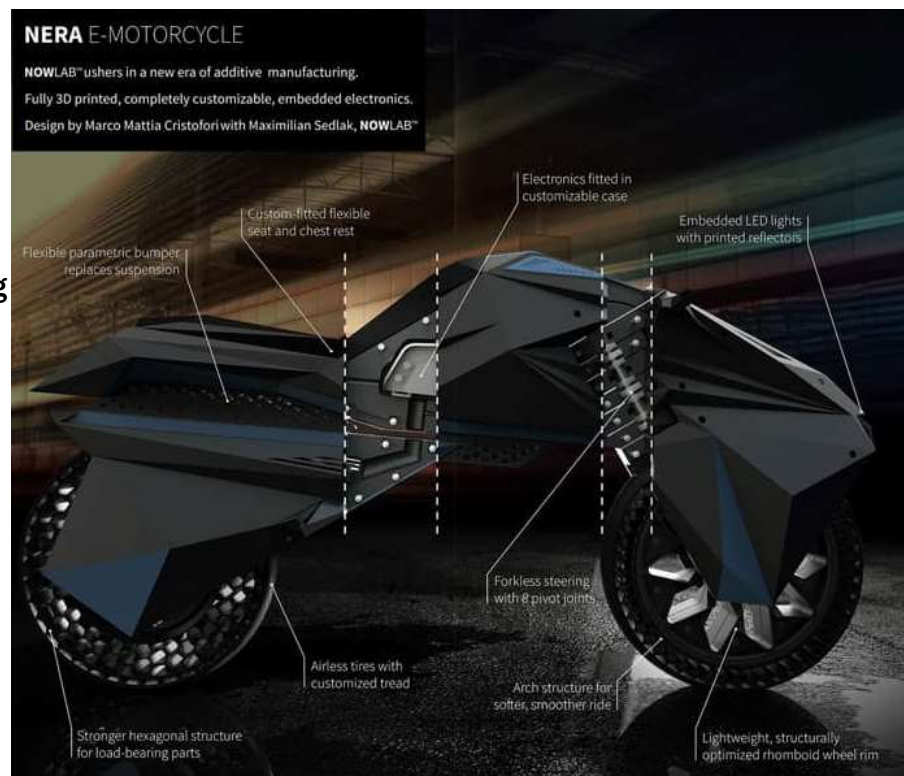
3D printing faces limitations such as the structural stability of printed objects.

Various 3D printer types: Fused Filament Manufacturing (FFM), Power and Glue, Stereo Lithography Apparatus (SLA), and Selective Laser Sintering (SLS),

Independent research

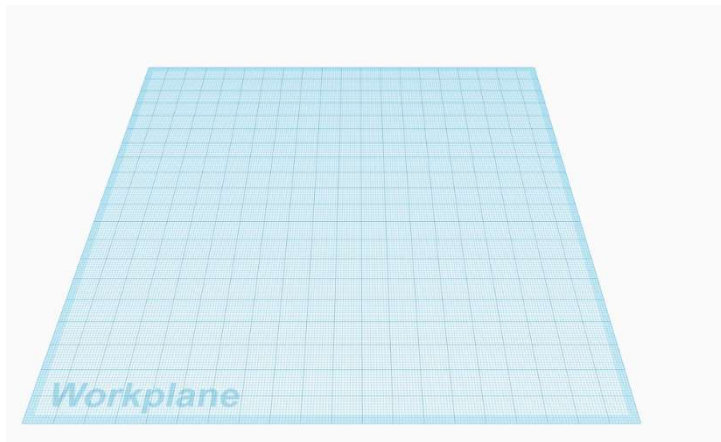
NERA. A new era in 3D printing.

3D printing's historical, artistic, and technological advancements have all lead up to the creation of the groundbreaking NERA E-Motorbike. Beginning with a discussion on digital evolution and Kodak's pivotal moment, the artistic expression in us all is what drives mathematical concepts, and technological advancements. The NERA E-Motorbike blossomed from this need for expression as a revolutionary, 3D-printed prototype, designed for large-format FFF printing. (PandaTranslator, 2024)



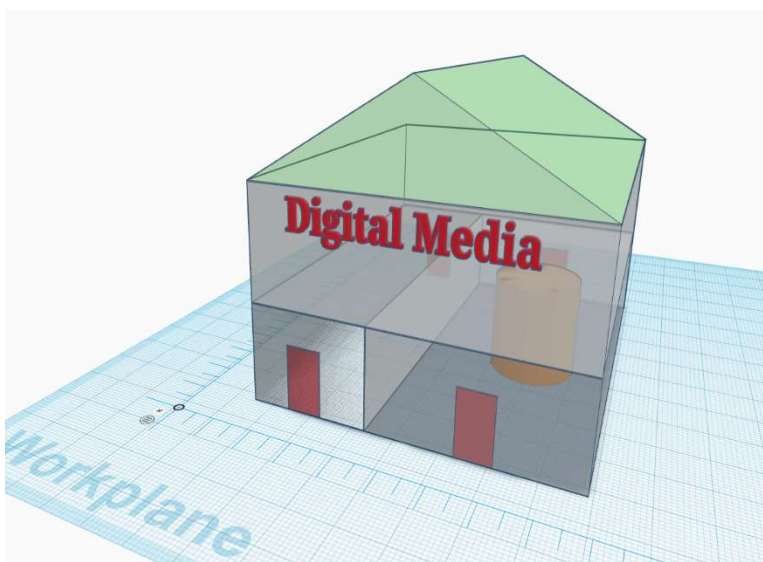
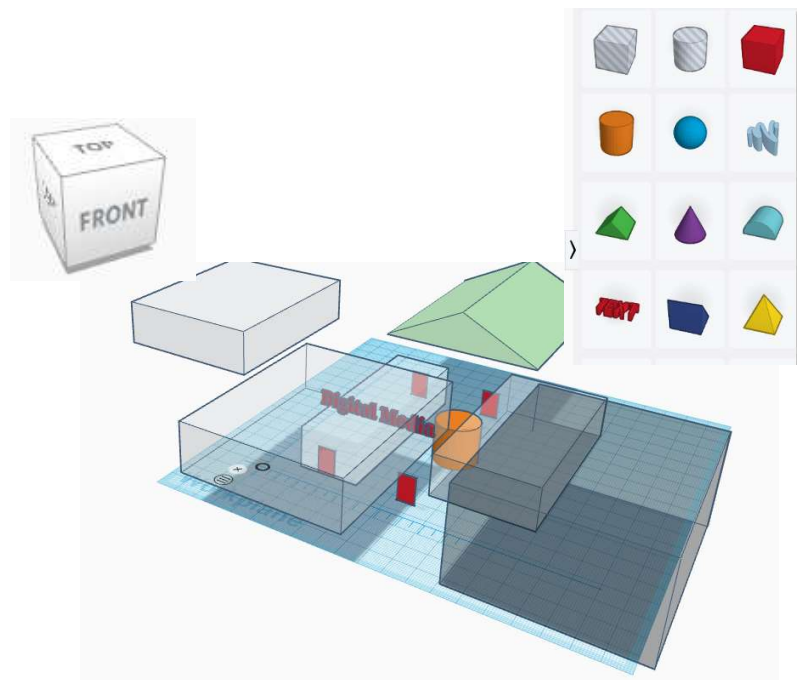
Beyond its visually striking design, the NERA showcases the adaptability and flexibility of 3D printing through non-printed embedded electronics and a modular system for individual adjustments. Research shows the NERA as a transformative force, illustrating the profound impact of 3D printing on traditional design paradigms and its potential to reshape mobility.

Practical - Tinkercad



I had a little experience with programs like CAD when it was first being taught in I.T at a GCSE level, unfortunately it has been a long time since then and am excited to learn more as these types of programs might be what I will need to become very versed in. I started with just a blank work plane as I could not load an SVG. File into the program, however, the task must be completed.

Using the objects provided by the program, I thought best to start simple and maybe build a house of some kind. Using the view block, various adjustment sliders on each object, using orientation via the work plane and colour/transparency options I attempted to fit the pieces together to make it look something like a structure. I found just messing about with the programs leads to fruitful discoveries.



I managed to assemble the pieces to fit well together. Like most of my ideas, my skill level leaves some to be desired, however it is a positive in the sense of I have a clear path for progression. I look forward to looking back on these one day and reminisce on my early days of skill development.

WEEK 10 – TYPOGRAPHY

Blue Pencil Summary:



Typography is the art of arranging and presenting text in a visually appealing and communicative way. Fonts, which are collections of different visual styles, are grouped into families, with Roman fonts being the most extensive, finding its foundation from the Latin alphabet. The historical progression of typography saw the introduction of Gothic type, known for its practicality but hindered readability. Aldus Manutius later innovated by introducing the 'Italic' type in the 1490s, aiming to strike a balance between style and legibility. In the world of typography, a character stands as an individual unit, representing a letter, number, or symbol. On the other hand, a glyph signifies the graphical shape within a character, such as the dot on the letter 'i.' Characters can incorporate multiple glyphs, and certain combinations, like ligatures such as 'fi,' contribute to the visual fullness and sophistication of fonts. This nuanced understanding of typography paved the foundation for effective design and communication practices.

Key Points:



Johannes Gutenberg's invention of the printing press in 1455.

Typeface is a collection of related fonts and fonts are stylized assembly of characters and symbols.

Roman fonts, play a significant role in organizing and categorizing fonts based on their origin from the Latin alphabet and stone inscriptions.

Independent research

Business Typography

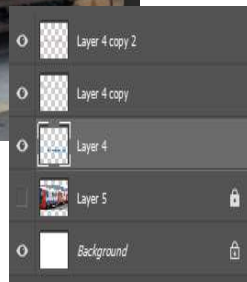
Business typography, also referred to as commercial typography, involves the strategic selection of fonts for commercial communication in both print and digital platforms. This practice employs various font types, such as Roboto, Serif, Sans Serif, Script, Helvetica, and Display fonts, each homing in on distinct design purposes (Kyle, 2021).



The choice of commercial typography is crucial for influencing message clarity, effectiveness, and overall visual appeal. It plays a significant role in design, contributing to the success of advertising campaigns by ensuring readability, avoiding confusion, and seamlessly adapting to different advertising goals. Typography's impact on design is very evident in its ability to convey clarity, boost attractiveness, and capture audience attention in advertising and sales. (Roca, 2024)

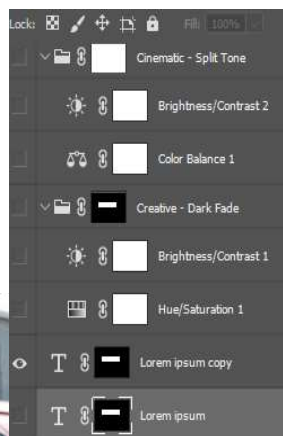
Practical - Text as art

I definitely like the idea of text as art, when I was presented with this task, I immediately thought of graffiti style of text. To me it was an original art text before digitized fonts. But my idea was it mix the old with the new so I found a stock photo of a train (common canvas of the graffiti artist)



Instead of graffiti I thought maybe use photoshop to add digital fonts in place of paint and use extraction and composition to layer the images together. I tried to used adjustments, colour gradients and filters to change the text slightly. It didn't come out as well as I had hoped but I believe spending more time with this I could achieve the desired outcome.

Three instances of the text 'Lorem ipsum' are shown in different styles: a yellow-to-purple gradient, a red-to-blue gradient, and a blue-to-white gradient.



The idea is doable, sadly I do not quite have the skill to make it work. However, here is a rendition of my idea, and again will be good to look back on this work in the future when my skills have far exceeded the level required for this composition.

Week 11 - Immersive Technology

Blue Pencil Summary:

Virtual Reality (VR) immerses users in a simulated environment using specialised headsets and haptic technology, shutting out the real world. Augmented Reality (AR) overlays digital information onto the real world through camera devices like glasses. Mixed Reality (MR/XR) combines aspects of both, offering an interactive experience.

VR enhances immersion with 360-degree images and videos, providing a lifelike experience. The incorporation of 3D graphics takes advantage of our binocular vision, creating depth through paired left images (predominantly blue wave) and right images (predominantly red). Various viewing methods, including polarising glasses, LCD shutter glasses, VR headsets, or anaglyph glasses, cater to 3D picture experiences.

Auditory enhancement is achieved through headphones, offering either surround sound or binaural 'Ambisonic' sound. These technologies together provide new and exciting ways to explore virtual environments, access augmented information, and seamlessly blend realities for a richer user experience.

Key Points:

Virtual Reality (VR) transports users to a virtual world, blocking out the real environment with headsets.

Integration of 3D graphics leverages binocular vision, creating depth through paired left/right image combinations.

The auditory dimension is crucial, delivered through headphones with surround sound or binaural 'Ambisonic' sound.

Independent research



Brain-Connected AR-VR Headset

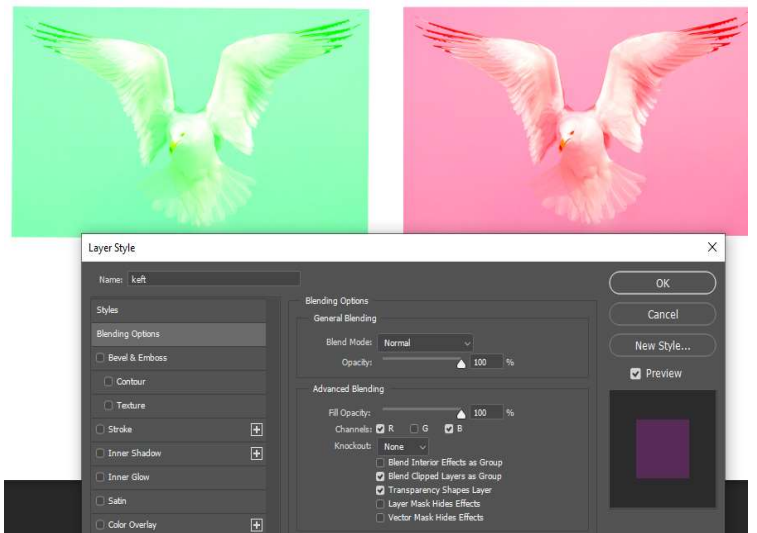
The future of Virtual Reality (VR) and Augmented Reality (AR) is evolving, with a shift from physical controllers to advanced inputs like hand- and eye-tracking. A notable player in this space is OpenBCI, a Brooklyn-based company developing non-invasive brain-computer-interface technology. OpenBCI's upcoming mixed-reality headset, Galea, integrates neural tech, employing EEG (electroencephalography) for brain signals, EMG (electromyography) for nerve and muscle activity, EDA (electrodermal activity) for stress-sensing, and PPG (photoplethysmography) for heart-rate monitoring. These sensors, combined with existing VR-AR headsets, offer exciting possibilities, including potential compatibility with platforms like Apple's Vision Pro. The convergence of neural technologies and mixed reality opens new avenues for immersive experiences. (Stein, 2023)

Practical – Anaglyphs



Anaglyphs are otherwise known as a base 3D image. Using the natural colour focus of each eye, anaglyphs are able to create images that can be viewed as 3D with special eyewear and equipment. I wanted to use an image that would appear as if it was flying out of the screen. I found this image and proceeded forward with it.

In order to achieve this, I made 3 copies of the image and named them appropriately. I then selected the left layer and used blending options to single out a colour channel. I did the same with the right image. Once colour blending was enabled I moved the images to slightly overlap giving the anaglyphic effect.



I do not have any 3D eyewear available at the time of this, however I am very confident based on other anaglyphic images I have seen that the image I have created here is in fact a successful attempt to create an anaglyph using Photoshop.

conclusion



During this course I have major evolutions in my understanding of digital media. The way we make art, build human connections, explore digital realms, find information, educate ourselves and in all ways that people experience life today, digital media plays a prominent role. From simple ideas like fonts to make text more fun and dynamic to crating physically explorable digital realms, digital media and technology has made great advancements in a short amount of time. I believe with more research, learning and skill development, I will achieve my goals of creating rich worlds, filled with memorable characters, telling thought provoking, emotive stories that the audience can live; that they can experience from a first-person perspective. The applications of Digital media, though already extensive, holds so much potential and I personally am excited to see where this path takes me, bestowing upon me new and exciting ways to leave a positive mark on the world.

L. Brighton

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